

Lab3 Written Justification of Component Diagram

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Architecture Selection: Layered Architecture

We chose Layered Architecture for the Vaccination Cohort & Dose Scheduling System. The system is organized into three main layers: the Presentation Layer, Business Logic Layer, and Data Layer. The Presentation Layer handles interactions such as citizen registration and vaccination scheduling, the Business Logic Layer handles eligibility checking, vaccination records and certificate generation, and the Data Layer stores the system's persistent information.

Reason 1 – Separation of Responsibilities:

Layered architecture separates user interaction from the core vaccination operations. Registration and scheduling functionality can remain in the Presentation Layer while operations such as dose eligibility checking, recording vaccinations, and generating certificates are handled by dedicated business components. This makes the system easier to understand, maintain, and modify.

Reason 2 – Centralized Data Management:

The system manages citizen details, vaccination history, scheduling information, and certificate information. Keeping these records in a dedicated Data Layer allows the business components to access information through defined interfaces rather than directly depending on the database implementation. This provides a consistent structure for managing vaccination data.

Security Advantage:

The layered structure helps protect sensitive citizen and vaccination information by separating the data layer from the presentation layer. Users interact with the system through controlled application interfaces rather than having direct access to the database. This provides a clear point at which access control and validation can be applied.

Performance Benefit:

Separating the business functions into components allows frequently used operations such as dose-eligibility checking and certificate verification to be handled independently. These operations can therefore be optimized without changing the presentation or data layers, while the clear separation of responsibilities avoids placing all processing in a single component.

Components: Citizen Registration, Vaccination Scheduling, Dose Eligibility, Vaccination Record, Certificate & QR, and Vaccination Database.